

Total No. of Questions : 8]

SEAT No. :

P-436

[Total No. of Pages : 2

[6003]-540

T.E. (AI & DS) (Semester - I)

PATTERN RECOGNITION

(317522 (B)) (2019 Pattern) (Elective - I)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q. 1 or Q. 2 , Q. 3 or Q. 4 , Q. 5 or Q. 6, Q. 7 or Q. 8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Discuss the Elements of Formal Grammars. [9]

b) Explain the Chomsky Normal form with suitable example. [8]

OR

Q2) a) Give an Abstract View of Parsing Problem. [9]

b) Describe Blocks Word Description String Generation example as Pattern Description. [8]

Q3) a) Differentiate between Homomorphism and Isomorphism. [9]

b) Draw and Explain Grammatical Interface Model and its objectives. [8]

OR

Q4) a) Describe Clique finding algorithm with suitable example. [8]

b) Explain the Design and Selection of Similarity Measures. [9]

P.T.O.

- Q5) a)** Describe Characteristics of Neural Computing Applications. **[9]**
- b) Explain CAM & other Neural Memory Structure. **[9]**

OR

- Q6) a)** Describe with neat diagram Artificial Neuron Activation and Output Characteristics. **[9]**
- b) Explain the different reasons to adopt a Neural Computational Architecture. **[9]**

- Q7) a)** Draw & Explain structure of a Multiple Layer Feedforward Network. **[9]**
- b) Explain Summary of the Back Propagation learning Procedure with suitable diagram. **[9]**

OR

- Q8) a)** How the character classification is done with Pattern Associator? **[9]**
- b) Draw & explain how to train the feedforward network using Generalized delta Rule? **[9]**

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